Design and Manufacture of an RC Aircraft

Overview:

This project comprised the complete conceptual design, aerodynamic analysis and physical manufacture of a fixed-wing RC aircraft, tasked with carrying payload while remaining stable and controllable using only electric propulsion. The aircraft must be modular and be capable of fitting into a cardboard transportation box, sized at 11200x660x433 mm.

The aircraft disassembled in the cardboard box is shown below:

Figure 9: Unfinished RC Aircraft disassembled in transportation box

An aircraft technical log was maintained, with the wing, tailplane, stability and performance parameters calculated. The aerodynamics were verified in ANSYS Fluent and the geometry was developed in SOLIDWORKS CAD before being manufactured using hot wire cut foam and plywood.

An initial cargo-style airframe was chosen as our strategy was to prioritise lifting capacity. However, this later had to be scrapped due to the limitations of the specified electric propulsion system and material yield strength. Thee design focus shifted towards a smaller, more aerodynamically efficient aircraft prioritising lift over velocity.

The airfoil selected was a standard unmodified Clark Y. This airfoil was selected for its favourable lift-to-drag ratio and simple manufacturability. The aircraft was driven by a 12x6 inch propeller using an E-Flite Power 10 (1100kV) motor and a single three-cell (2200 mAh) lithium-polymer battery. MTOW was 1.6 kg with 350 g allocated for payload.

Figure 10: Finished RC Aircraft undergoing bench testing

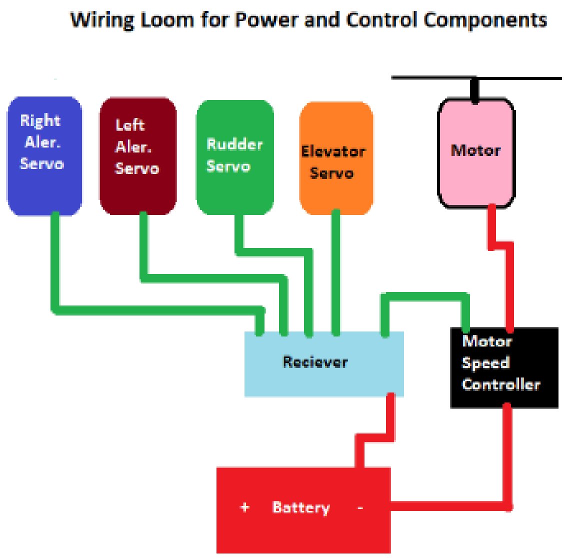


Figure 11: Wiring Loom for Power and Control Components

Longitudinal stability was addressed through the sizing of the components and the placement of the centre of gravity. The neutral point was located at 45.3% of the MAC aft of the leading edge; with the loaded centre of gravity at ~ 25% of the chord. This resulted in a static margin of ~ 0.22. A tailplane setting angle of 2.99° relative to the horizontal fuselage datum was chosen to trim the aircraft in cruise, accounting for a downwash gradient of 0.53 and $\frac{dC_L}{d\eta}$ = 0.85.

The centre of gravity was subsequently verified experimentally using a three-point weighing strategy, with the measured centre of gravity aligning with the predicted centre of gravity derived from the SOLIDWORKS model.

The wing was sized to a gross area of 0.275m^2 at an aspect ratio of 4.4, resulting in a MAC of 0.25m. The lift-curve slope for the 3D Clark Y was calculated to be 4.19 per radian, and the maximum lift coefficient was 1.39. The Reynolds number was 1.95x10^5 and Oswald efficient factor was obtained as 0.921.

Table 2: Velocity Performance Parameters of Aircraft

Performance Parameter	Value (m/s)
Stall Speed	8.56
Take-Off Speed	10.27
Cruise Speed	11.12
Maximum Speed	13.82

Two-dimensional CFD analysis was performed in ANSYS Fluent to determine the aerodynamic characteristics. The determined C_L and C_D (Coefficient of Lift and Drag respectively) aligned with the values in the tech log, verifying the turbulence model and meshing strategy.

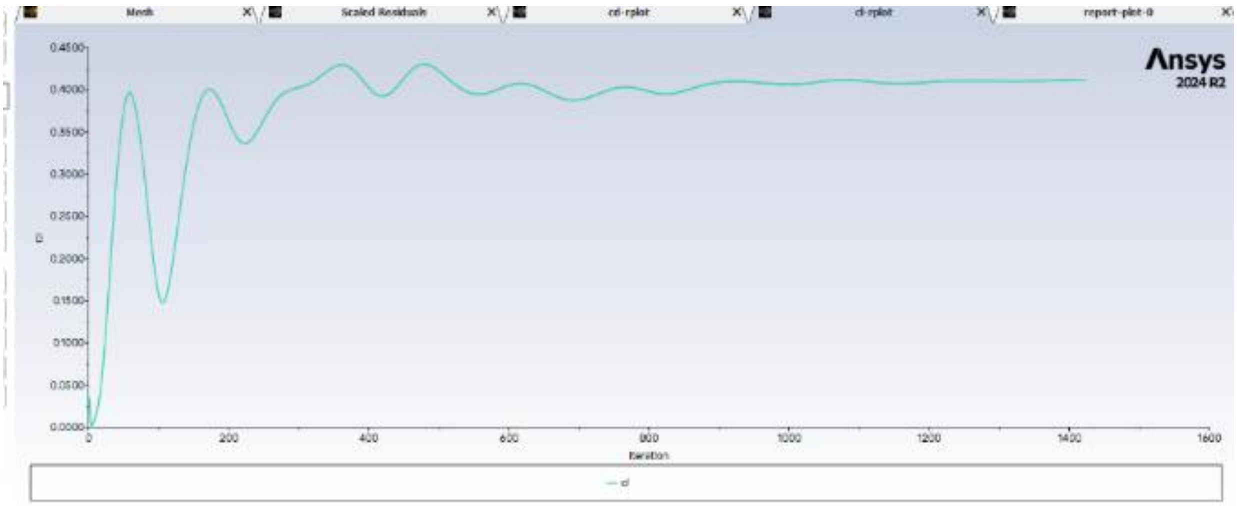


Figure 12: C_L of Airfoil

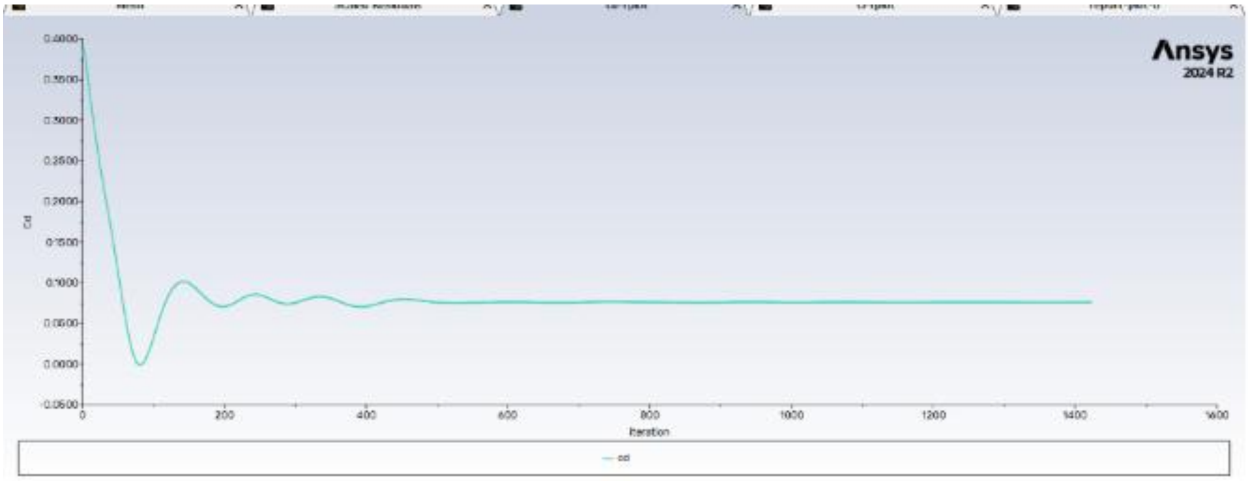
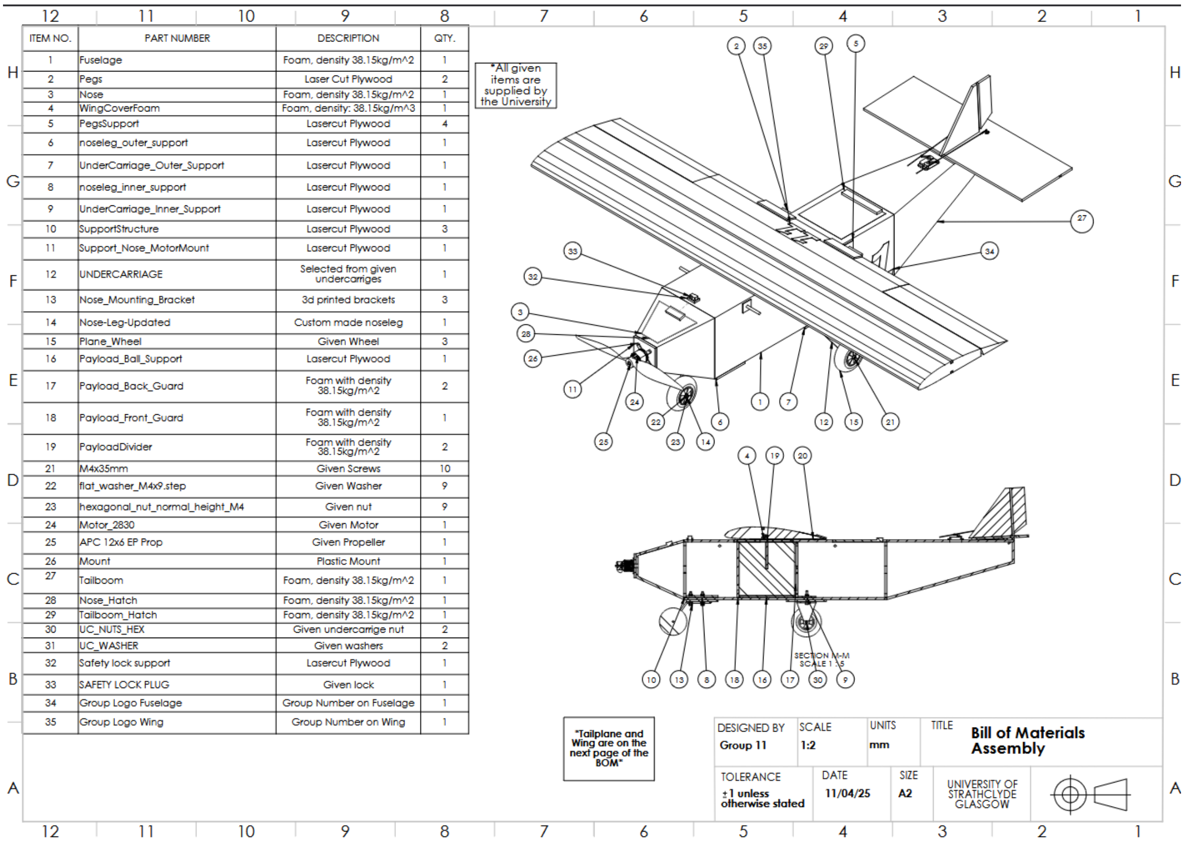


Figure 13: C_D of Airfoil

The aircraft was manufactured principally from 5mm foamboard, with the wing and control surfaces hot wire cut from foam and plywood supports DXF laser cut. The fuselage assembly was joined using riveted joints, which improved structural robustness when compared to joining individual panels.

A payload compartment of four interlocking foamboard slots was designed to retain the cargo (tennis balls) and restrict payload movement. The nose-leg mounts were 3D-printed to suit the airframe, while a proprietary main undercarriage was adopted. The control. The foam aerofoil was reinforced with a plywood spar maximise structural integrity while minimising added weight. The control system was wired with the aileron servo leading to a channel cut into the underside of the wing to preserve a clean aerodynamic surface, which can be seen in ***Figure 9.*** Servo function and motor response were verified with a bench test, ensuring the aircraft is ready for test flight. The aircraft underwent a successful test flight with the payload despite considerable turbulence arising from weather conditions.

Technical drawings were prepared outlining the BOM and the manufacturing techniques required to assemble the aircraft.



ITEM NO.	PART NUMBER	DESCRIPTION	QTY.
1	Fuselage	Foam, density 38.15kg/m^2	1
2	Pegs	Laser Cut Plywood	2
3	Nose	Foam, density 38.15kg/m^2	1
4	WingCoverFoam	Foam, density: 38.15kg/m^3	1
5	PegsSupport	Lasercut Plywood	4
6	noseleg_outer_support	Lasercut Plywood	1
7	UnderCarriage_Outer_Support	Lasercut Plywood	1
8	noseleg_inner_support	Lasercut Plywood	1
9	UnderCarriage_Inner_Support	Lasercut Plywood	1
10	SupportStructure	Lasercut Plywood	3
11	Support_Nose_MotorMount	Lasercut Plywood	1
12	UNDERCARRIAGE	Selected from given undercarriges	1
13	Nose_Mounting_Bracket	3d printed brackets	3
14	Nose-Leg-Updated	Custom made noseleg	1
15	Plane_Wheel	Given Wheel	3
16	Payload_Ball_Support	Lasercut Plywood	1
17	Payload_Back_Guard	Foam with density 38.15kg/m^2	2
18	Payload_Front_Guard	Foam with density 38.15kg/m^2	1
19	PayloadDivider	Foam with density 38.15kg/m^2	2
21	M4x35mm	Given Screws	10
22	flat_washer_M4x9.step	Given Washer	9
23	hexagonal_nut_normal_height_M4	Given nut	9
24	Motor_2830	Given Motor	1
25	APC 12x6 EP Prop	Given Propeller	1
26	Mount	Plastic Mount	1
27	Tailboom	Foam, density 38.15kg/m^2	1
28	Nose_Hatch	Foam, density 38.15kg/m^2	1
29	Tailboom_Hatch	Foam, density 38.15kg/m^2	1
30	UC_NUTS_HEX	Given undercarrige nut	2
31	UC_WASHER	Given washers	2
32	Safety lock support	Lasercut Plywood	1
33	SAFETY LOCK PLUG	Given lock	1
34	Group Logo Fuselage	Group Number on Fuselage	1
35	Group Logo Wing	Group Number on Wing	1

Figure 14: BOM of Aircraft

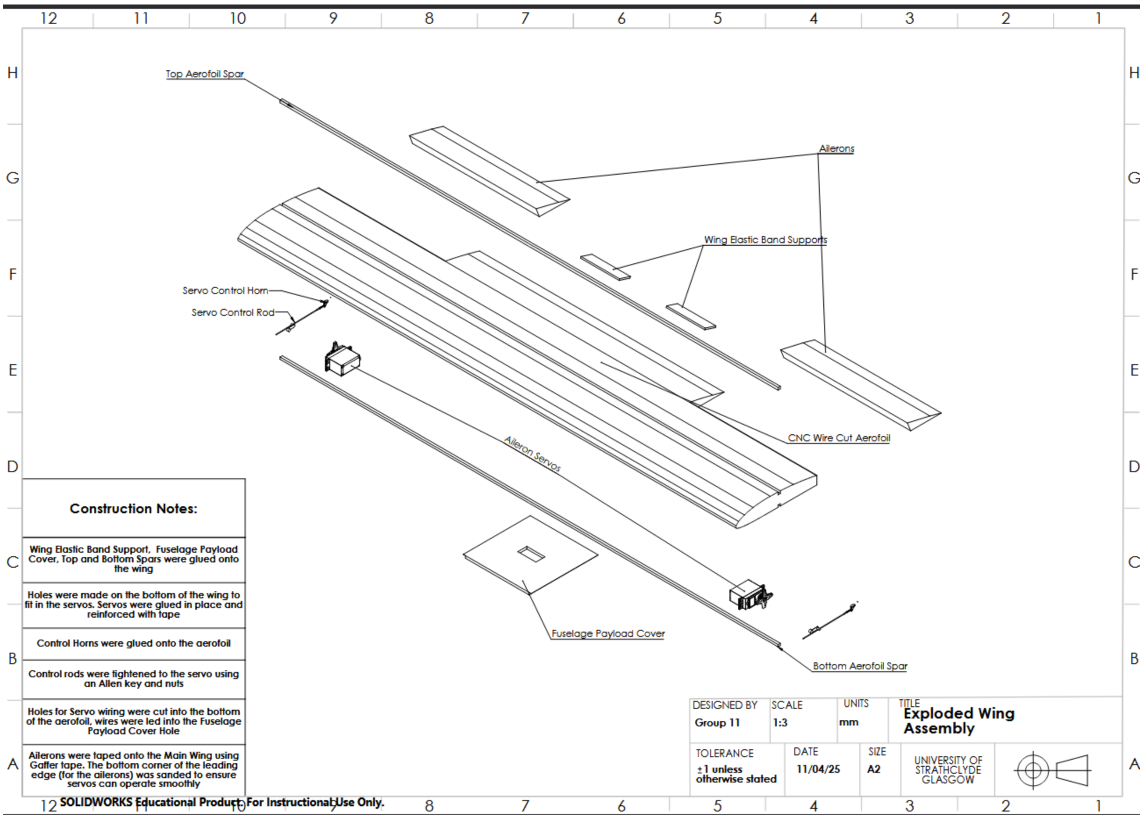


Figure 15: Wing Assembly Drawing